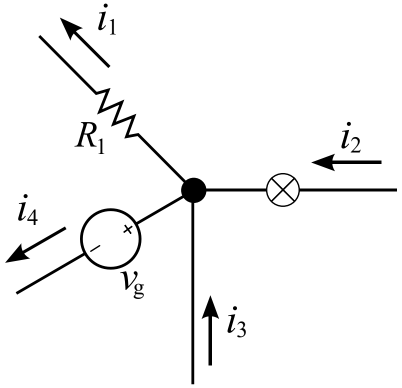
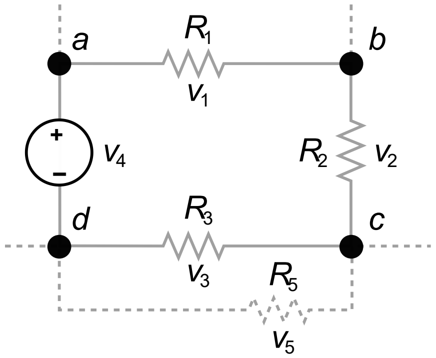
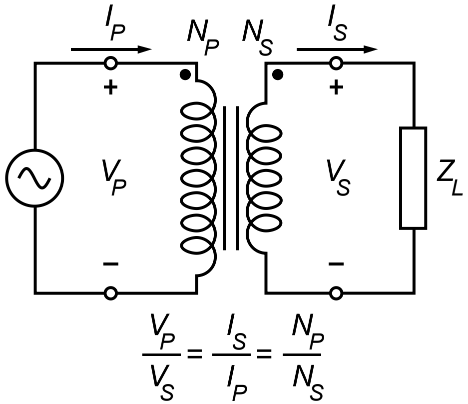
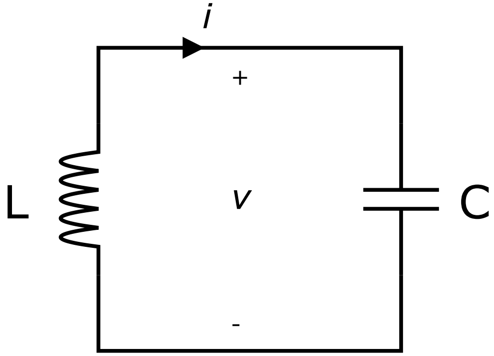
Hochschule Kaiserslautern
University of Applied Sciences

Introduction to Electrical Engineering









Overview

1. Physical Principles and Basic Concepts
2. Direct Current Circuits
3. **Electric and Magnetic Fields**
4. Alternating Current Technology
5. Wrap-Up and Outlook

Electric and Magnetic Fields

- Coulomb Force
- Electrostatic Induction
- Capacitance & Capacitor
- Magnetic Force
- Electromagnetic Induction
- Inductivity & Inductor
- Transformer
- Maxwell‘s Equations
- Transient Responses

Coulomb Force

vector field associating to each point in space the Coulomb force experienced by a unit test charge

for point charges Q:

$$\vec{F}_2 = \vec{e}_r \frac{1}{4\pi\varepsilon_0} \frac{Q_1}{r^2} \cdot Q_2 = \vec{E}_1 \cdot Q_2$$

$$\vec{F}_1 = -\vec{F}_2$$

strives to reduce potential difference between charged bodies

→ no electrostatic field in (ideal) conductors

Electric Field

$$\vec{E} = \vec{e}_r \frac{Q}{4\pi\varepsilon_0 r^2}$$

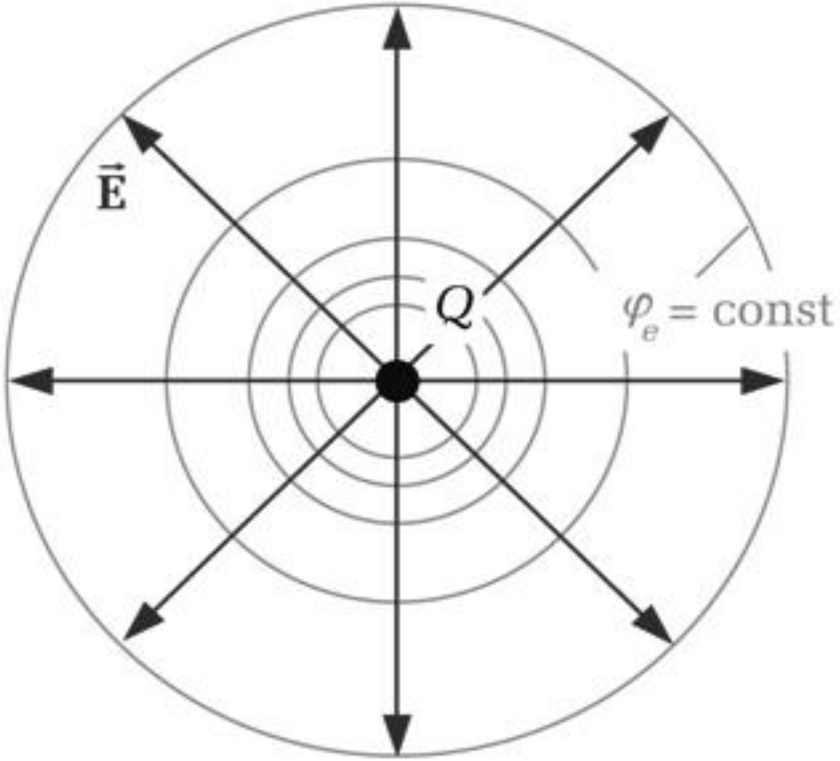
$$[E] = 1\ \frac{\mathrm{V}}{\mathrm{m}}$$

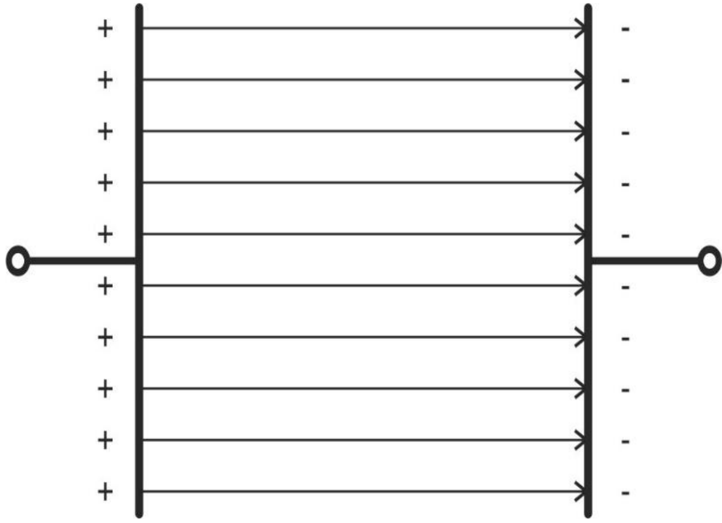
electrical field strength

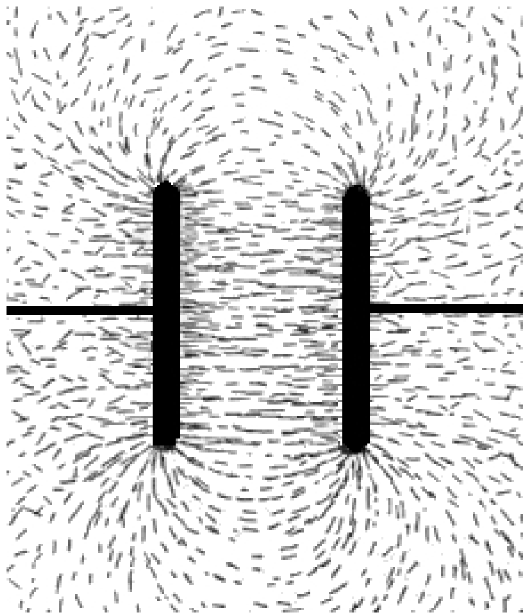
Electric Field Lines

- field lines follow direction of electric field
- start at positive charges and end at negative charges
- density indicates field strength
- equipotential lines/surfaces perpendicular to electric field lines

point charge:

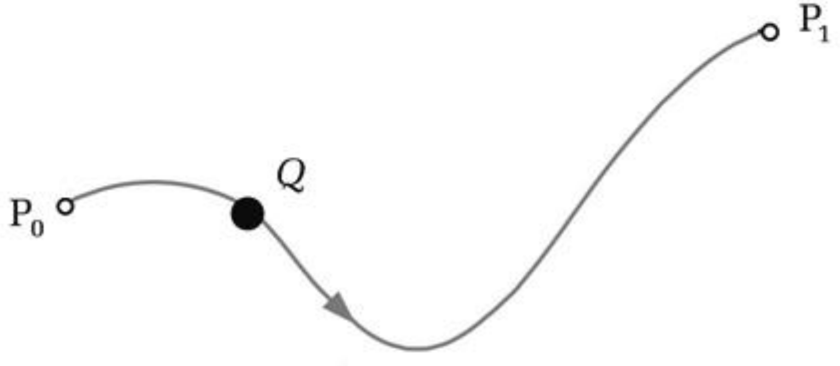






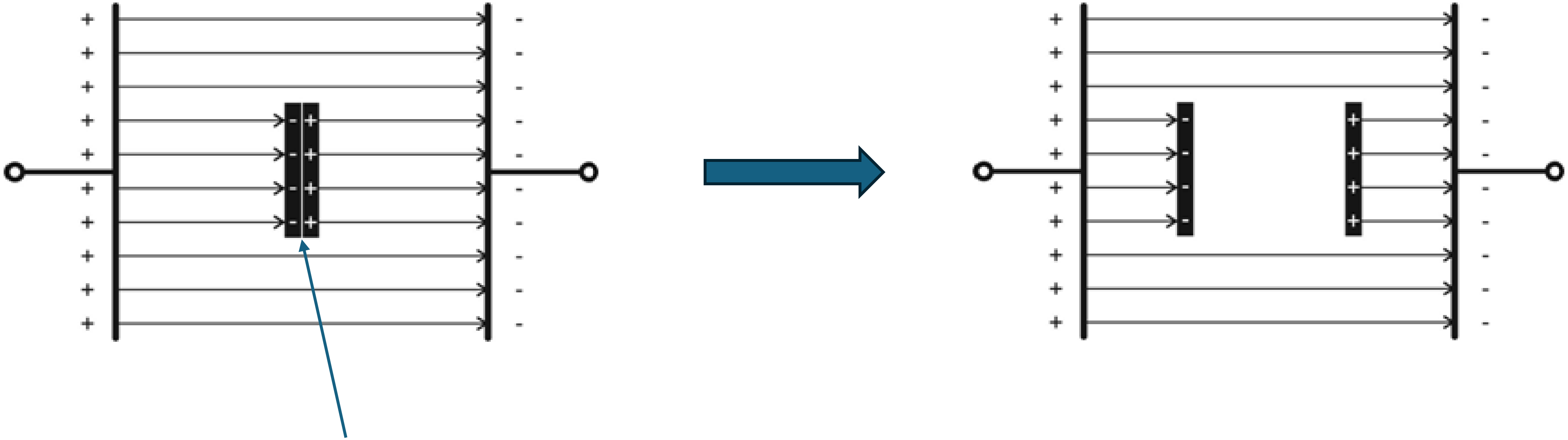
Voltage and Potential

$$\varphi_1 = -\int_{P_0}^{P_1} \vec{E} \cdot d\vec{l}$$



$$V_{12} = \varphi_1 - \varphi_2 = -\int_{P_0}^{P_1} \vec{E} \cdot d\vec{l} + \int_{P_0}^{P_2} \vec{E} \cdot d\vec{l} = \int_{P_1}^{P_2} \vec{E} \cdot d\vec{l}$$

Electrostatic Induction



conductor in electric field
→ electrostatic induction, aka influence
(caused by Coulomb force)

induced surface charges cancel
electric field in conductor
→ Faraday cage

Electric Displacement

$\vec{D} = \varepsilon \vec{E}$

vacuum permittivity
(physical constant)

$\varepsilon_0 = 8{,}854 \cdot 10^{-12} \, \frac{\mathrm{As}}{\mathrm{Vm}}$

polarization

$D = \varepsilon_0 E + P$

absolute permittivity: $\varepsilon = \varepsilon_0 \cdot \varepsilon_r$

relative permittivity,
aka dielectric constant
(material dependent)

$[D] = 1 \, \frac{\mathrm{As}}{\mathrm{m}^2}$

D abstracts away (absorbs) polarization effects (bound charges displaced by induction) from E, leaving a field that couples only to free charges.

→ D is an auxiliary field! E is the actual, physical field.

Dielectric Constants

insulators (dielectrics) in electric field:

electrons slightly shifted relative to nuclei (polarization), the more the higher ε_r

→ weakens electric field inside the dielectric

	ε_r
air	≈ 1
water	≈ 80
glass	$\approx 5-10$
plastics / polymers	$\approx 2-4$
ferroelectric ceramics	up to several thousand

inside perfect conductors: $D = 0$ (since $E = 0$), although $\varepsilon_r \to \infty$ (free charges)

Gauss‘s Law

surface integral over electric displacement yields sum of enclosed charges:

$$\oiint \vec{D}\, d\vec{A} = Q_{\mathrm{free}}$$

free charges only
(polarization included in D)

differential form (equivalent to integral form):

$$\nabla \cdot \vec{D} = \rho_{\mathrm{free}}$$

divergence

charge density

Plate Capacitor

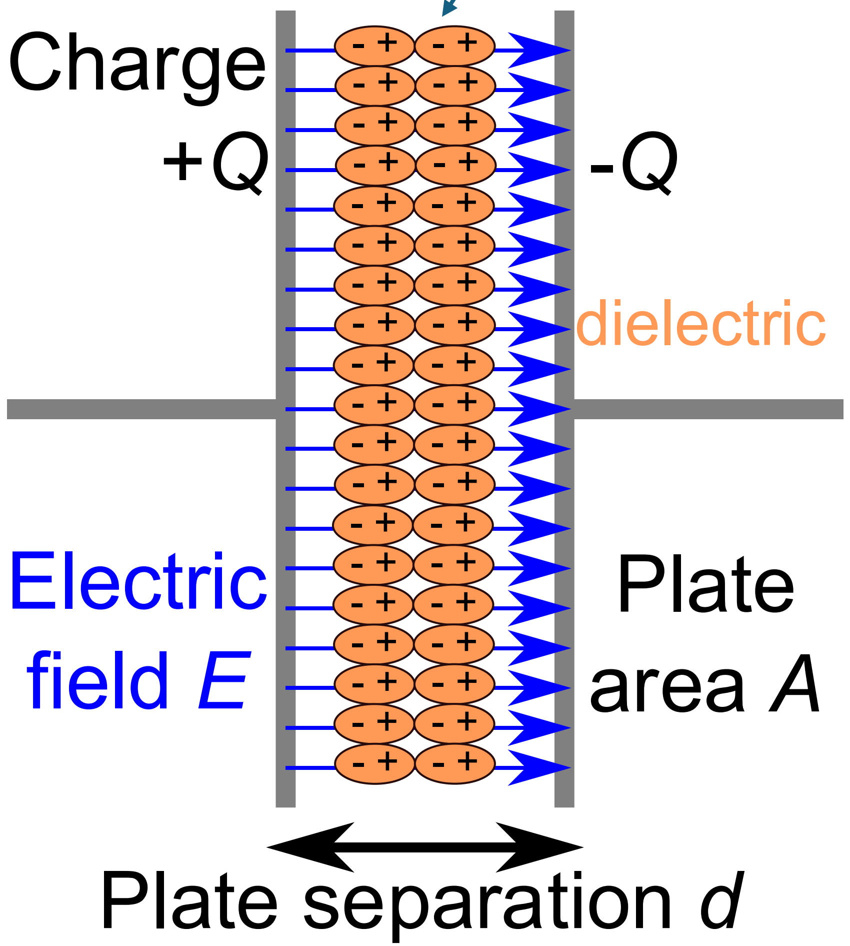
$$D = \frac{Q}{A}$$

$$\varepsilon_0 \cdot \varepsilon_r \cdot E \cdot d = \frac{Q}{A} \cdot d$$

$$\underbrace{\frac{\varepsilon_0 \cdot \varepsilon_r \cdot A}{d}}_{C} \cdot V = Q$$

(depends only on geometry and dielectric)

polarization counteracts electric field
→ more charge can be stored



Capacitance

measure of a system’s ability to store electric charge, for a given potential difference

$$C = \frac{Q}{V}$$

$$[C] = 1\,\frac{\mathrm{As}}{\mathrm{V}} = 1\mathrm{F} \quad \text{(farad)}$$

Time Dependence

$$Q = C \cdot V$$

$$\frac{dQ}{dt} = C \cdot \frac{dv}{dt}$$

$$i = C\,\frac{dv}{dt}$$

no DC through capacitors

voltage across capacity is continuous (since charge cannot jump infinitely fast)

current through capacity can be discontinuous (because it is the derivative of the voltage)

Energy of Capacitance

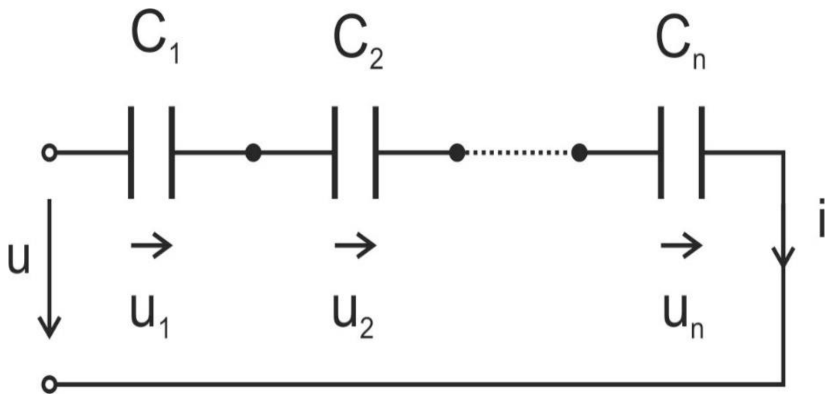
capacitor as energy storage:

$$W = \int i \cdot v \, dt = \int C \frac{dv}{dt} \cdot v \, dt = C \int v \, dv = \frac{1}{2} C v^2$$

Capacitors in Series

$$V = V_1 + V_2 + \cdots + V_n$$

$$V = \frac{Q}{C_1} + \frac{Q}{C_2} + \cdots + \frac{Q}{C_n}$$



$$v = \frac{1}{C_1}\int i\,dt + \frac{1}{C_2}\int i\,dt + \cdots + \frac{1}{C_n}\int i\,dt$$

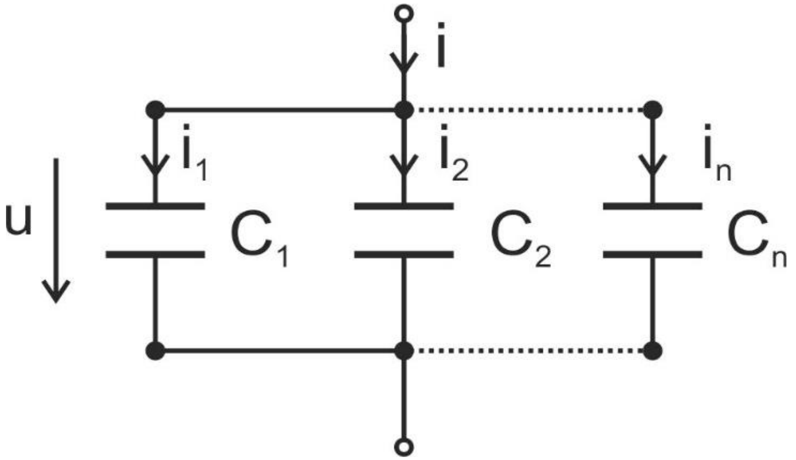
$$\boxed{\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n} = \sum_{k=1}^{n}\frac{1}{C_k}}$$

for two capacitors: $C = \frac{C_1 C_2}{C_1+C_2}$; $\quad \frac{v_1}{v_2} = \frac{C_2}{C_1}$; $\quad v_1 = v\frac{C_2}{C_1+C_2}$ (voltage divider rule)

Capacitors in Parallel

$$Q = Q_1 + Q_2 + \cdots + Q_n$$

$$Q = C_1 V + C_2 V + \cdots + C_n V$$



$$i = C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} + \cdots + C_n \frac{dv}{dt}$$

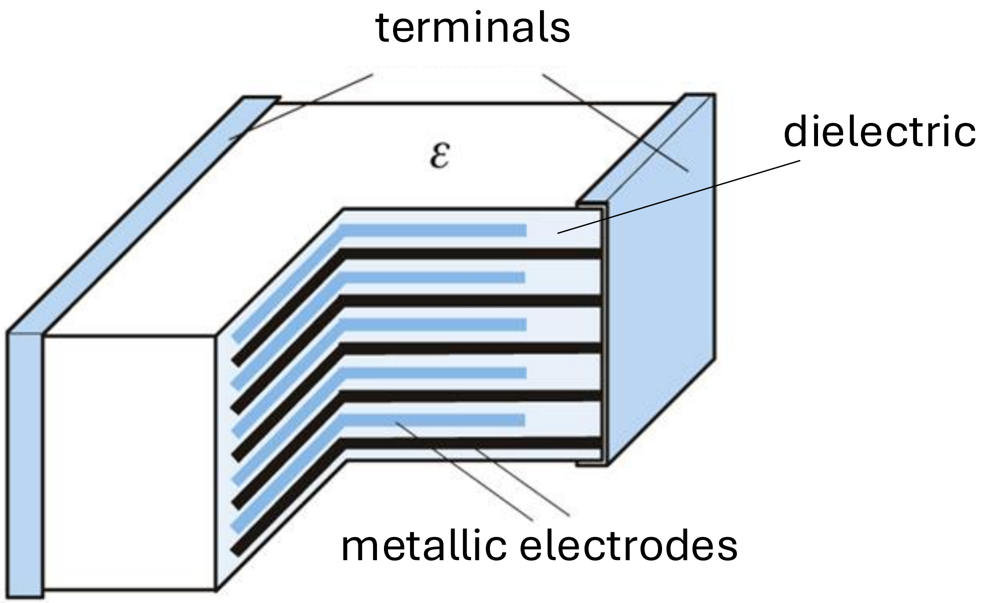
$$\boxed{C = C_1 + C_2 + \cdots + C_n = \sum_{k=1}^{n} C_k}$$

current divider rule: $i_k = i \frac{C_k}{C}$

for two capacitors: $\frac{i_1}{i_2} = \frac{C_1}{C_2}$

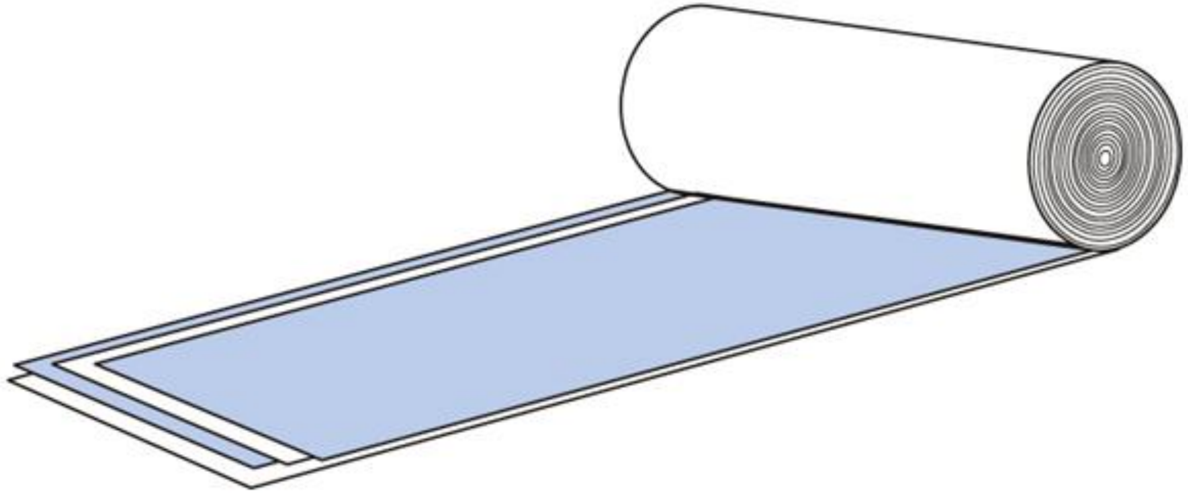
Common Implementations

connected in parallel → higher capacitance



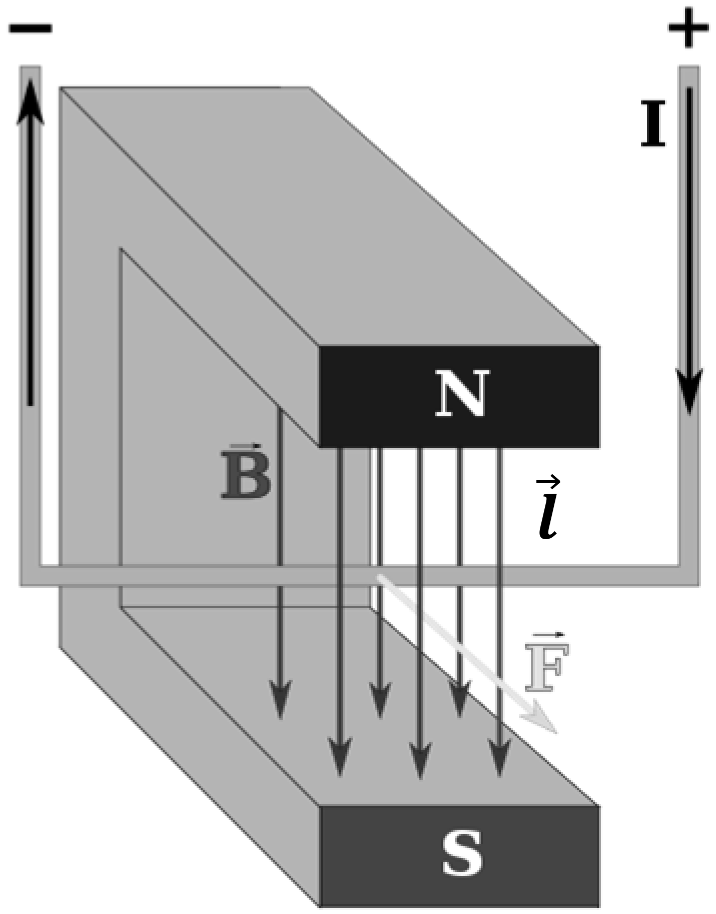
multilayer ceramic capacitor (MLCC)

larger area → higher capacitance



wound capacitors (film or electrolytic)

Magnetic Force



a magnet exerts a force on a current-carrying conductor:

$$F \sim I \cdot l \quad \rightarrow \quad F = B \cdot I \cdot l$$

magnetic field

Magnetic Flux Density

$$[B] = 1\ \frac{\mathrm{N}}{\mathrm{Am}} = 1\ \frac{\mathrm{Vs}}{\mathrm{m}^2} = 1\ \mathrm{T}\ \text{(tesla)}$$

also: $1\ \mathrm{T} = 10{,}000\ \mathrm{G}$ (gauss)

direction of magnetic force is orthogonal to plane of conductor and field, strength depends on angle between conductor and field:

$$\vec{F} = I \cdot \vec{l} \times \vec{B}$$

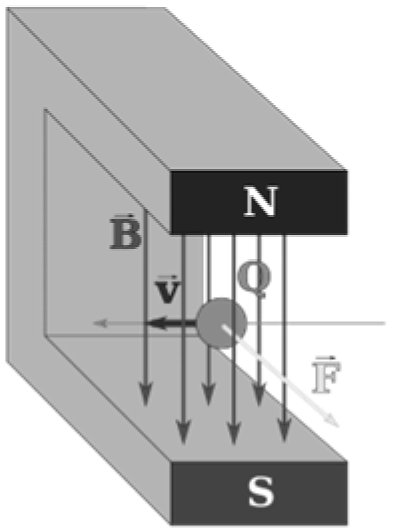
$$F = \left|\vec{F}\right| = B \cdot I \cdot l \cdot \sin\alpha$$

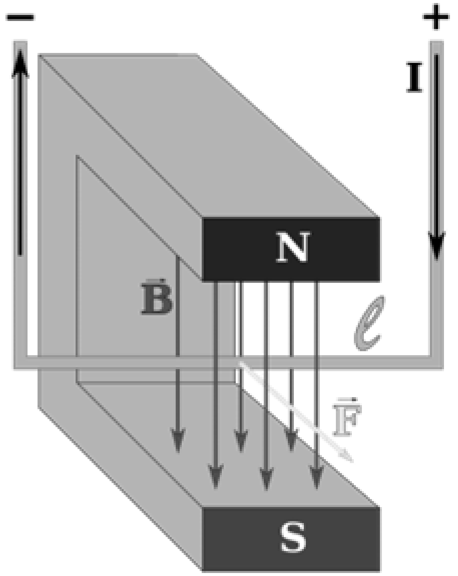
Lorentz Force

same kind of magnetic force on charged particles:

$$\vec{F} = I \cdot \vec{l} \times \vec{B} = Q \cdot \vec{v} \times \vec{B}$$

including electric field: $\vec{F} = Q \cdot \left(\vec{E} + \vec{v} \times \vec{B}\right)$

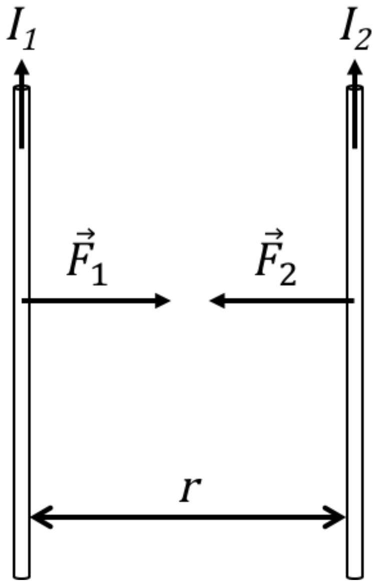




Generation of Magnetic Fields

- magnetic field exerts force on moving electric charges
- in turn, moving electric charges generate magnetic fields (in permanent magnets: orbital and spin motion of electrons within atoms)

Ampère's force law



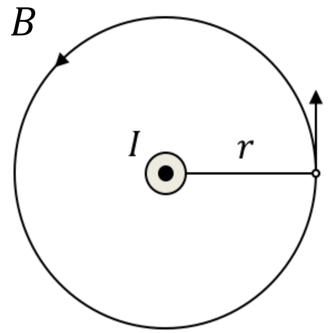
$$F_2 = I_2 \cdot B_1 \cdot l$$

$$B_1 = \frac{\mu_0 I_1}{2\pi r}$$

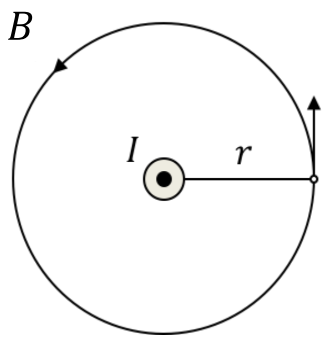
circumference

$$\mu_0 = 4\pi \cdot 10^{-7} \ \frac{\mathrm{Vs}}{\mathrm{Am}}$$

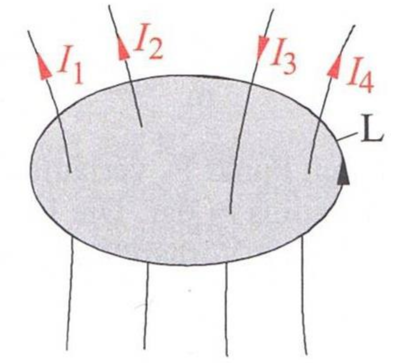
vacuum magnetic permeability
(physical constant)



Ampère's Circuital Law



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$



$$I = I_1 + I_2 - I_3 + I_4$$

differential form (equivalent to integral form):

$$\nabla \times \vec{B} = \mu_0 \vec{J}$$

rotation

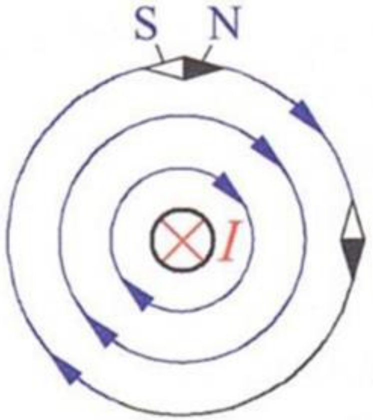
current density

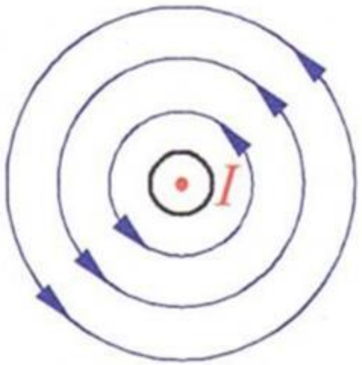
only closed circuits in nature: magnetic fields look like a dipole from the outside

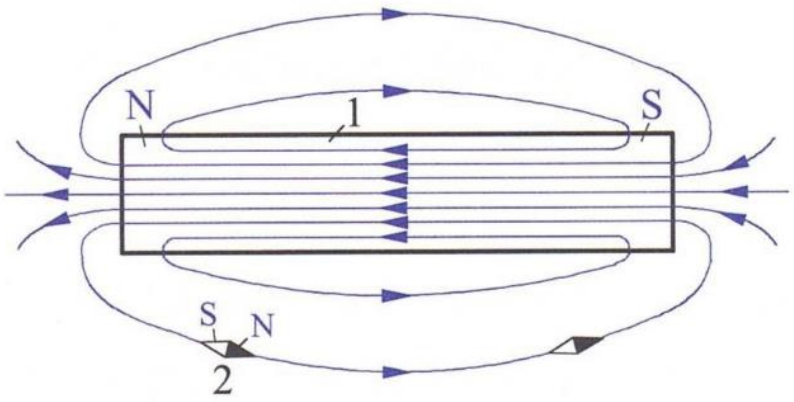
Magnetic Field Lines

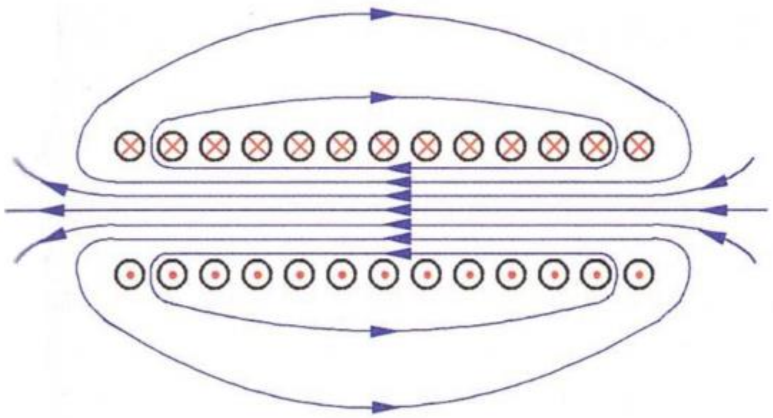
closed field lines from north to south pole (dipoles, no magnetic monopoles)

right-hand corkscrew rule: circular field lines around a conductor

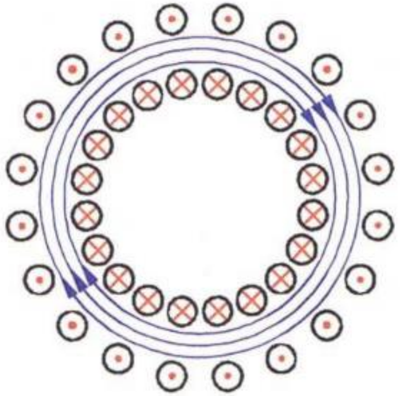








electromagnetic coil



Magnetic Field Strength

magnetization

$$H = \frac{1}{\mu_0} B - M$$

$$\vec{B} = \mu \cdot \vec{H}$$

vacuum permeability

permeability: $\mu = \mu_0 \cdot \mu_\mathrm{r}$

relative permeability

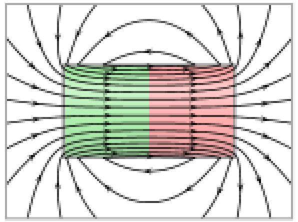
(material dependent, > 1000 for ferromagnetic materials)

$$[H] = 1\ \frac{\mathrm{A}}{\mathrm{m}}$$

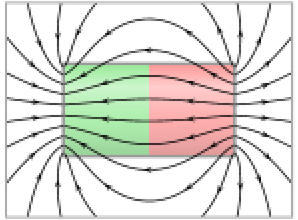
H abstracts away (absorbs) magnetization effects (bound currents) from B, leaving a field that couples only to free currents.

→ H is an auxiliary field! B is the actual, physical field.

(beware of confusion from naming and notation conventions)



$\vec{B}$



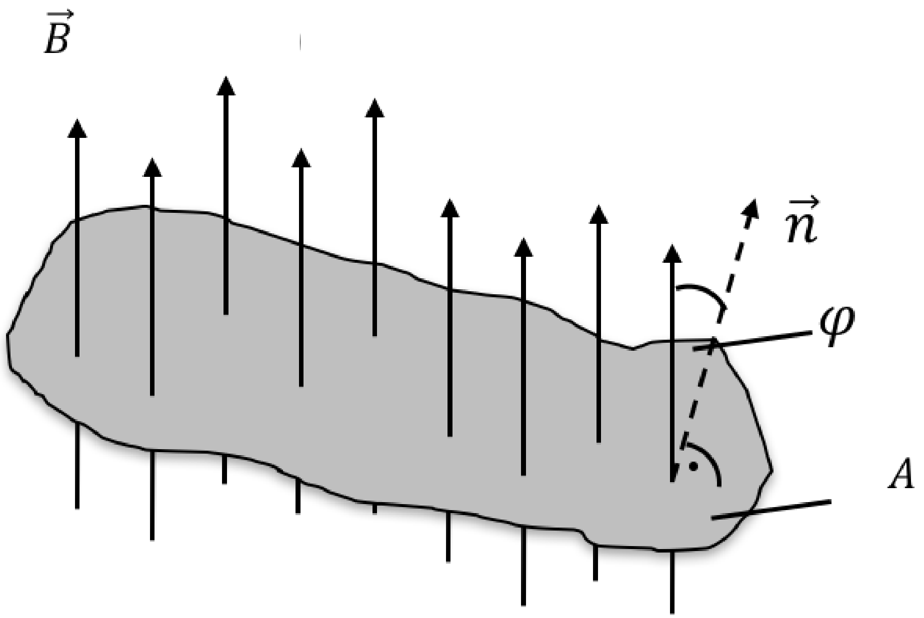
$\vec{H}$

$\vec{M}$

Magnetic Flux

$$\Phi = \vec{B} \cdot \vec{A}_n = B \cdot A \cdot \cos\varphi$$

$[\Phi] = 1\,\mathrm{Vs} = 1\,\mathrm{Wb}$ (weber)



Gauss‘s Law for Magnetism

magnetic flux through **closed** surface is zero (no magnetic monopoles):

$$\oiint \vec{B}\, d\vec{A} = 0$$

differential form (equivalent to integral form):

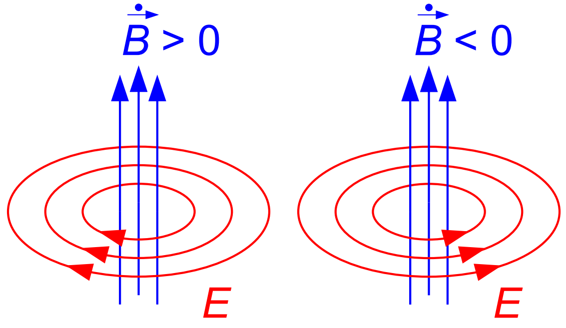
$$\nabla \cdot \vec{B} = 0$$

(The magnetic flux through an **open** surface need not be zero.)

Electromagnetic Induction

changing magnetic field (through surface) induces current in conductor

Faraday's Law of electromagnetic induction:



$$V = -\frac{d\Phi}{dt}$$

voltage induced

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt}\int \vec{B}\, d\vec{A}$$

$$\nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$$

induced voltage always opposes the change that caused it (Lenz's law)

Inductance

old: 

measure of a system's ability to store magnetic energy (in its magnetic field), for a given electric current

$$L = \frac{\Phi}{I}$$

$$[L] = 1\,\frac{\mathrm{Vs}}{\mathrm{A}} = 1\mathrm{H} \quad \text{(henry)}$$

Time Dependence

$$I = \frac{\Phi}{L}$$

$$\frac{d\Phi}{dt} = L \cdot \frac{di}{dt}$$

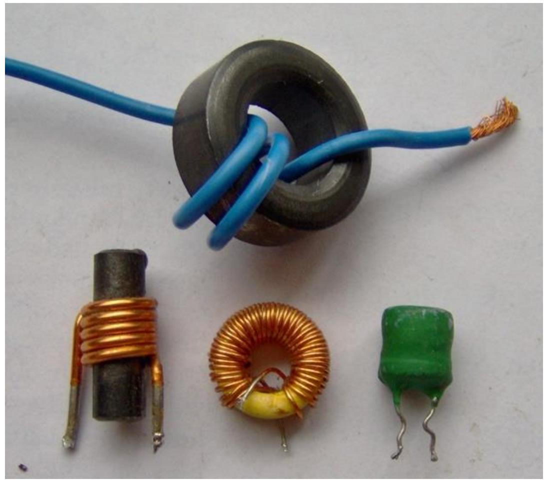
$$v = L\frac{di}{dt}$$

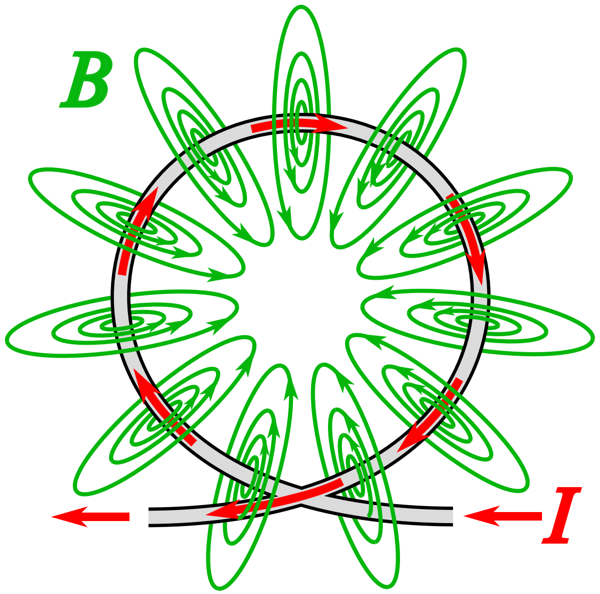
inductor is short circuit for DC

current through inductance is continuous

voltage across inductance can be discontinuous

Inductor (aka Coil)

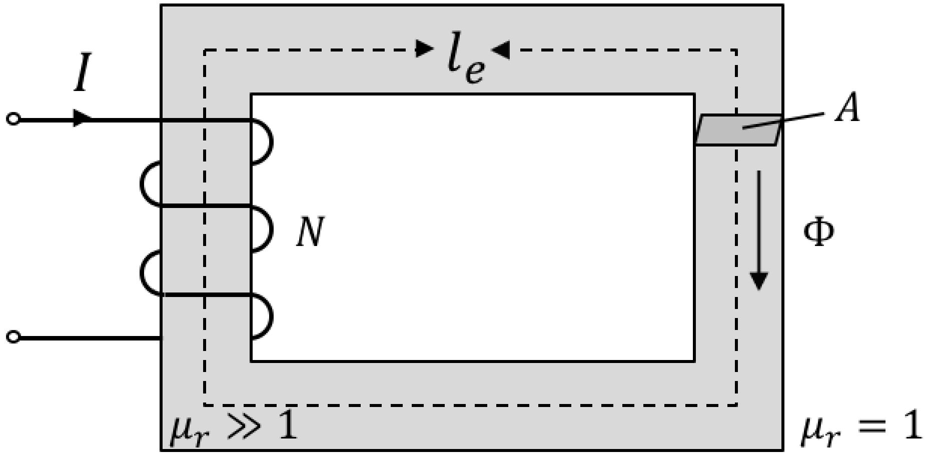




- magnetic effect of a coil depends on product of current and number of turns, expressed as ampere-turns: NI
- inductance measures how effectively a coil links the current with the magnetic flux through its core (flux linkage in weber-turns): $L = \frac{N\Phi}{I}$

Magnetic Analogue of Ohm's Law

ampere-turns create magnetic flux through the core of a coil (like voltage pushes current), depending on magnetic reluctance R_m of core/path:



$$\Phi = B \cdot A = \frac{\mu \cdot N \cdot I}{l} \cdot A \quad \text{with} \quad R_m = \frac{l}{\mu \cdot A}$$

length and cross-sectional area of the magnetic path, not the wire

$$\boxed{\Phi = \frac{NI}{R_m}}$$

$$\rightarrow L = \frac{N\Phi}{I} = \frac{N^2}{R_m}$$

Electric vs Magnetic Circuits

electrons moving in conductor, no electric field

magnetic field, no moving particles

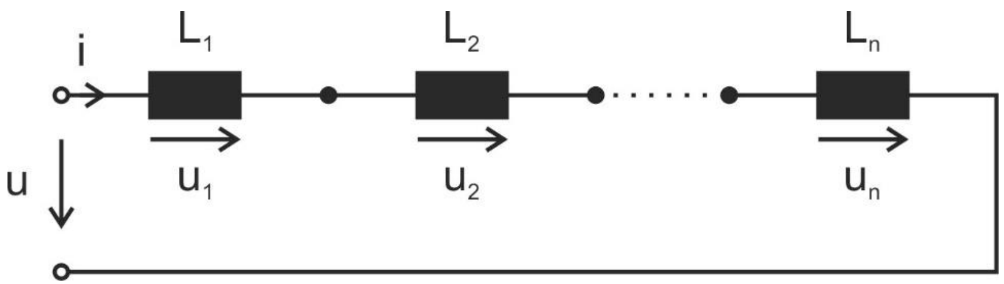
	Electric Network	Magnetic Network
	$R = \frac{\varrho l}{A}$	$R_m = \frac{l}{\mu A}$
Ohm:	$V = RI$	$NI = R_m \phi$
KVL:	$\sum_{\text{loop}} RI = 0$	$\sum_{\text{loop}} R_m \phi = \sum NI$
KCL:	$\sum_{\text{node}} I = 0$	$\sum_{\text{node}} \phi = 0$

Energy of Inductance

inductor as magnetic energy storage:

$$W = \int i \cdot v \, dt = \int i \cdot L \frac{di}{dt} dt = L \int i \, di = \frac{1}{2} L i^2 = \frac{1}{2} (Ni) \Phi$$

Inductors in Series



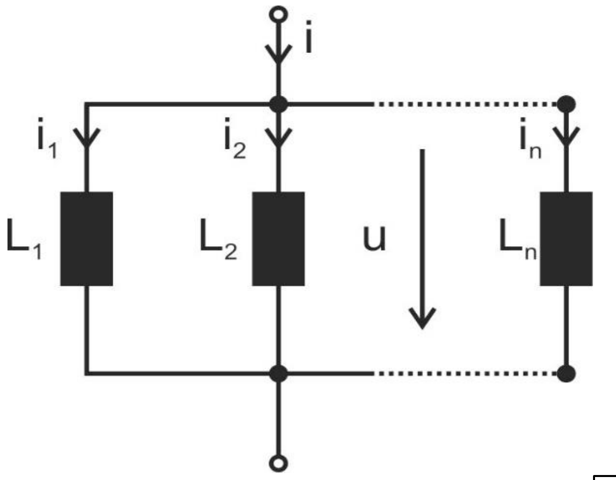
$$v = L_1 \frac{di}{dt} + L_2 \frac{di}{dt} + \cdots + L_n \frac{di}{dt}$$

$$\boxed{L = L_1 + L_2 + \cdots + L_n = \sum_{k=1}^{n} L_k}$$

voltage divider rule: $v_k = v \frac{L_k}{L}$

for two inductors: $\frac{v_1}{v_2} = \frac{L_1}{L_2}$

Inductors in Parallel

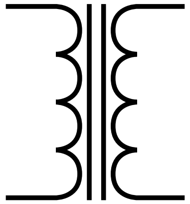


$$i = \frac{1}{L_1}\int v\,dt + \frac{1}{L_2}\int v\,dt + \cdots + \frac{1}{L_n}\int v\,dt$$

$$\boxed{\frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2} + \cdots + \frac{1}{L_n} = \sum_{k=1}^{n} \frac{1}{L_k}}$$

for two inductors: $L = \frac{L_1 L_2}{L_1 + L_2}$; $\frac{i_1}{i_2} = \frac{L_2}{L_1}$; $i_1 = i\,\frac{L_2}{L_1 + L_2}$ (current divider rule)

Transformer

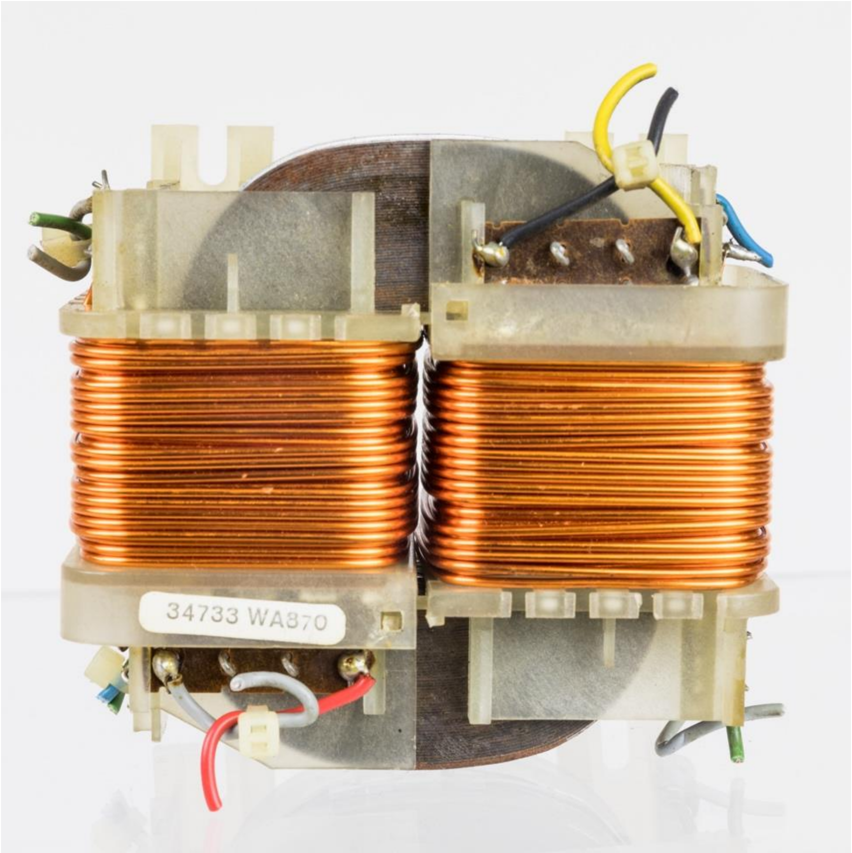


transfers electrical energy from one circuit into another

works through electromagnetic induction between two coils

coils magnetically coupled via a common core (shared magnetic flux), without any direct conductive connection

essential for transmission of electric power (in the form of alternating current)



Ideal Transformer

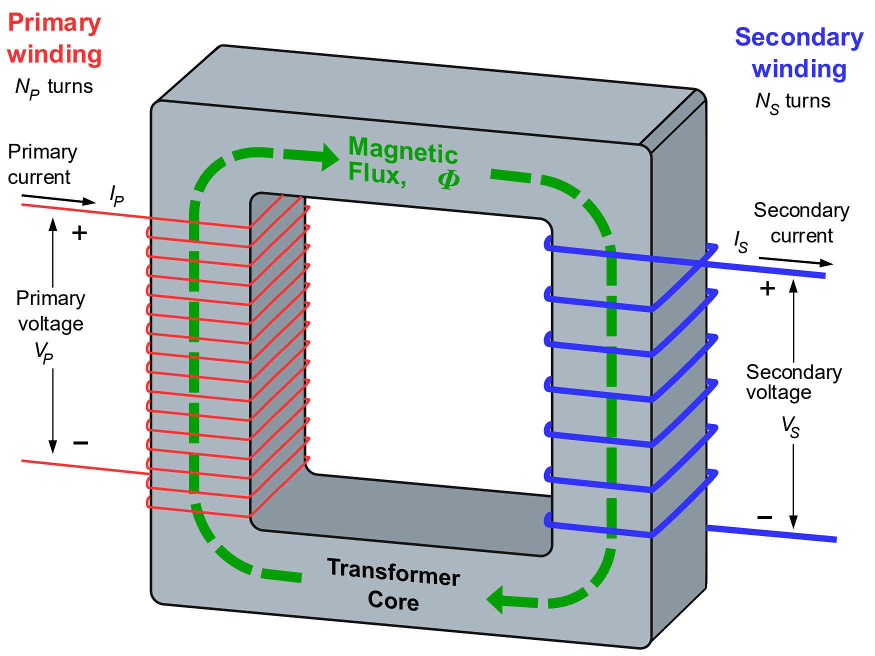
$$V_P = -N_P \frac{d\Phi}{dt} \qquad V_S = -N_S \frac{d\Phi}{dt}$$

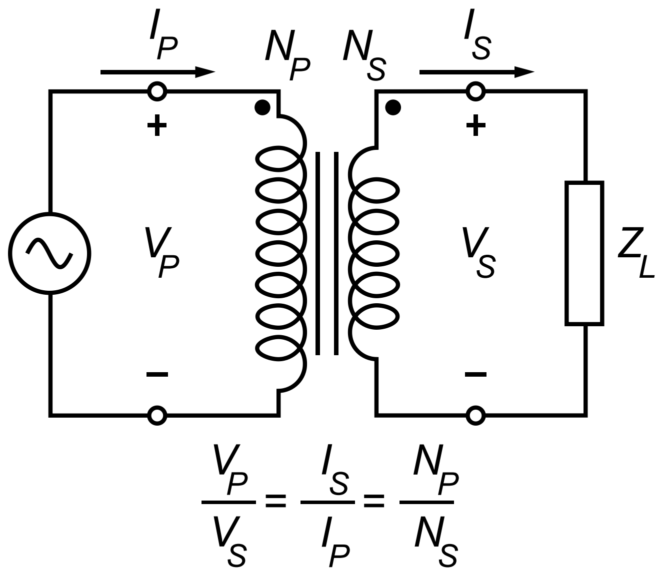
$$\rightarrow \frac{V_P}{V_S} = \frac{N_P}{N_S} = a$$

$a > 1$: step-down transformer

$a < 1$: step-up transformer

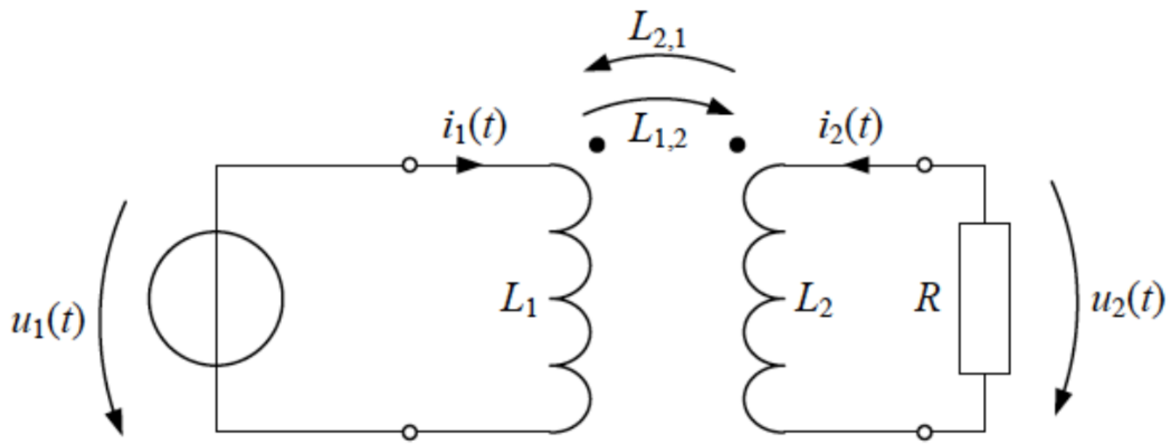
higher voltage → less heat loss during energy transfer ($P_{\text{loss}} = U_R \cdot I = I^2 \cdot R$ and $P = U_S \cdot I$)





Load on Secondary Side

need to consider mutual inductance:



$$u_1 = L_1 \frac{di_1}{dt} + L_{2,1} \frac{di_2}{dt}$$

$$u_2 = L_2 \frac{di_2}{dt} + L_{1,2} \frac{di_1}{dt} = -i_2 \cdot R$$

with $\quad L_{1,2} = L_{2,1} = \frac{N_P \cdot N_S}{R_m}$

Maxwell‘s Correction to Ampère‘s Law

symmetry to electromagnetic induction:
a time-varying electric field produces a circulating magnetic field

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I \boxed{+ \mu_0 \varepsilon_0 \frac{d}{dt} \int \vec{E}\, d\vec{A}}$$

displacement current

$$\nabla \times \vec{B} = \mu_0 \vec{J} \boxed{+ \mu_0 \varepsilon_0 \frac{d\vec{E}}{dt}}$$

$\frac{\partial B}{\partial t}$ → rotational E, $\frac{\partial E}{\partial t}$ → rotational B:
→ electromagnetic waves in vacuum (no charges or wires needed)

Maxwell‘s Equations in Vacuum

integral form:

- $\oiint \vec{E}\, d\vec{A} = \frac{Q}{\varepsilon_0}$
- $\oiint \vec{B}\, d\vec{A} = 0$
- $\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B}\, d\vec{A}$
- $\oint \vec{B} \cdot d\vec{l} = \mu_0 I + \mu_0 \varepsilon_0 \frac{d}{dt} \int \vec{E}\, d\vec{A}$

differential form:

- $\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$
- $\nabla \cdot \vec{B} = 0$
- $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$
- $\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$

wave equations:

$$\nabla^2 \vec{E} - \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2} = 0$$

$$\nabla^2 \vec{B} - \frac{1}{c^2} \frac{\partial^2 \vec{B}}{\partial t^2} = 0$$

$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$ (speed of light in vacuum)

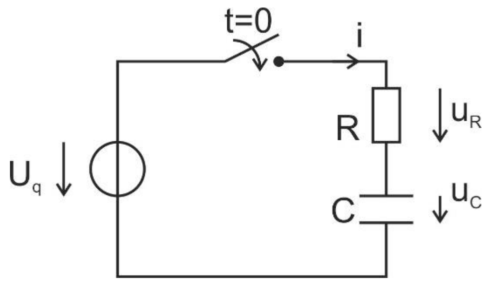
Transient Responses: RC Series Circuit

transients: time-varying currents and voltages resulting from the sudden application of sources (usually due to switching)

differential equation:

$$U_q = iR + \frac{1}{C}\int i\,dt$$

charging the capacitor with DC:

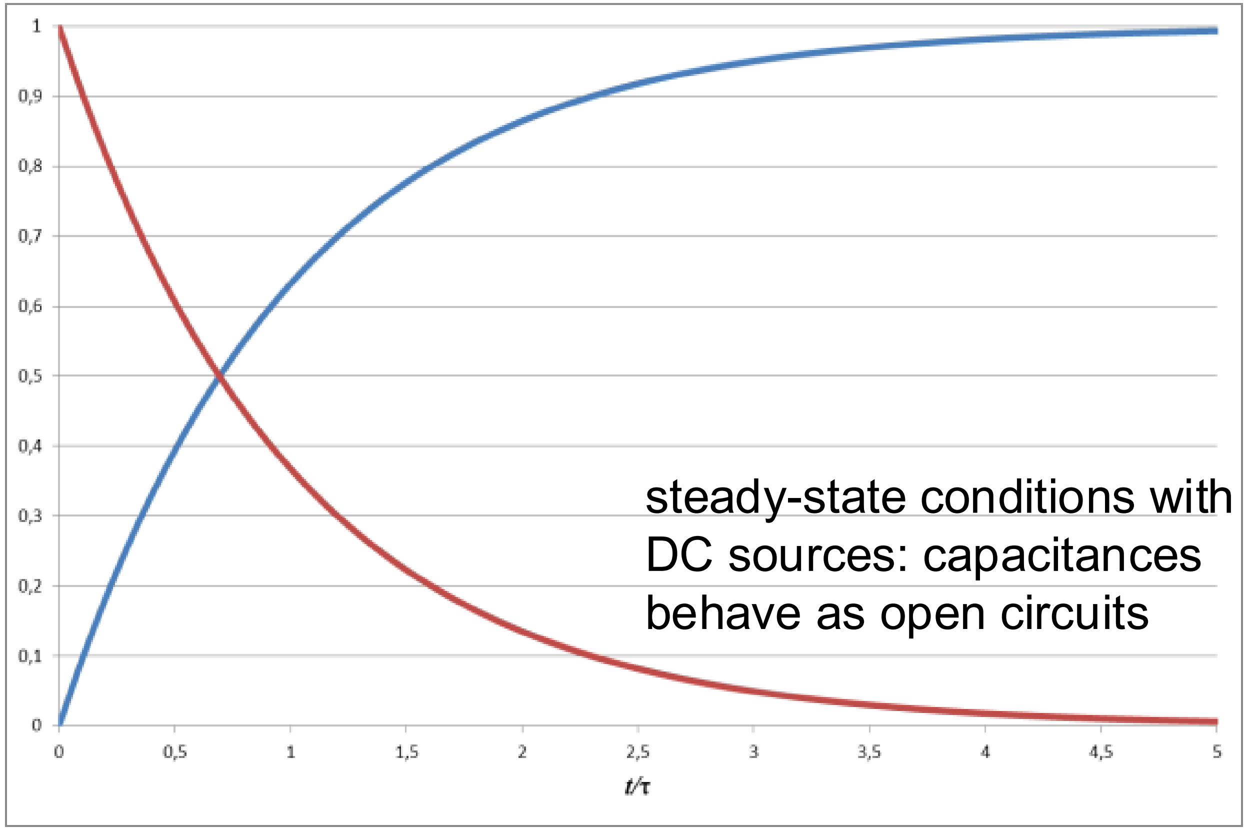


$$i = \frac{U_q}{R} \cdot e^{-\frac{t}{\tau}}$$

$$\tau = RC$$

$$u_R = U_q \cdot e^{-\frac{t}{\tau}}$$

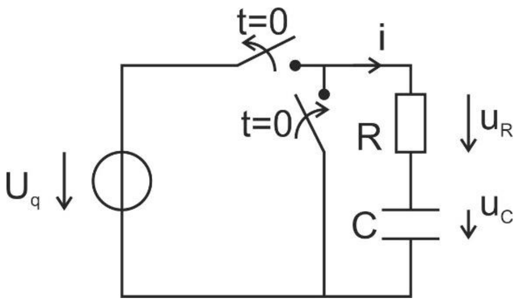
$$u_C = U_q\left(1 - e^{-\frac{t}{\tau}}\right)$$

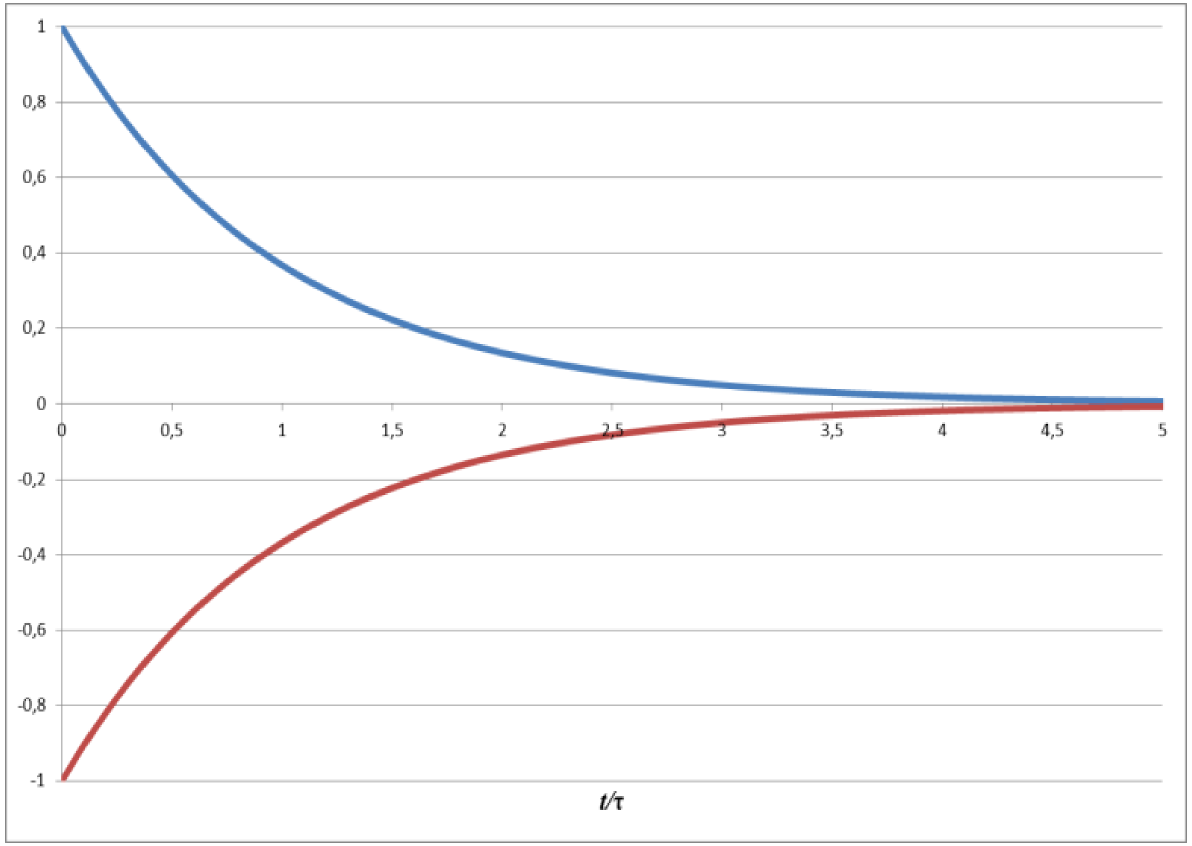


steady-state conditions with DC sources: capacitances behave as open circuits

Transient Responses: RC Series Circuit

discharging the capacitor:

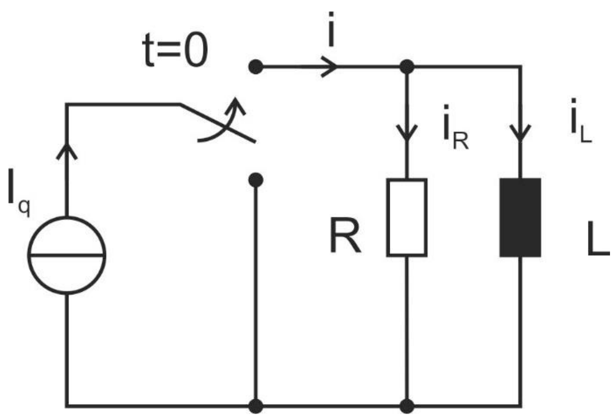




Transient Responses: RL Parallel Circuit

differential equation:

$$I_q = \frac{u}{R} + \frac{1}{L}\int u\,dt$$



$$u = I_q R \cdot e^{-\frac{t}{\tau}}$$

$$\tau = \frac{L}{R}$$

$$i_R = I_q \cdot e^{-\frac{t}{\tau}}$$

$$i_L = I_q\left(1 - e^{-\frac{t}{\tau}}\right)$$